ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-SAMPLE) **SUMMER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 20**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. What is the definition of a convex set? 5 (CO1) (PO1)

Solution:

A set C is convex if for any two points $x, y \in C$ and any $t \in [0, 1]$, the point $tx + (1 - t)y$ also lies in C. In plain terms, the straight line segment between any two points in the set must lie entirely within the set.

Rubric:

- 5 points for correct definition.

2. In Bayes' Theorem, what do the terms prior, likelihood, and posterior refer to? 5 (CO1) (PO1)

Solution:

Prior P(H) — our belief about the hypothesis before seeing data. Likelihood P(D|H) — probability of observing the data given the hypothesis. Posterior P(H|D) — updated belief about the hypothesis after observing the data.

Rubric:

- 1.67 points for correctly stating each segment.

3. Explain why convexity is a desirable property for a machine learning cost function. 5 (CO1) (PO1)

Solution:

A convex cost function guarantees that any local minimum found is also the global minimum. Gradient descent will reliably converge to the best possible solution regardless of the starting point. When the cost function is non-convex, gradient descent can get stuck at local minima or saddle points, meaning the model may converge to a suboptimal solution depending on initialization.

Rubric:

- 5 points for correctly explaining.

4. For a univariate linear regression model $h_\theta(x) = \theta_0 + \theta_1 x$ with training data $\{(1,1),(2,2),(3,3)\}$, compute $J(\theta)$ when $\theta_0 = 0$ and $\theta_1 = 0.5$. 5 (CO1) (PO1)

Solution:

$J(\theta) = \frac{1}{2\times 3}(0.25 + 1 + 2.25) = 0.583$

Rubric:

- 5 points for correctly finding the loss.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-B) **SUMMER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. For a simple dataset $\{(1, 1.75), (2, 3.22), (3, 1.67)\}$, calculate one step of gradient update, if $h_\theta(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$ and initially, $\theta_0 = 0, \theta_1 = 0$ and $\alpha = 1$. 10 (CO1) (PO1)

Solution:

Gradient for θ_0: $\frac{\partial J(\theta)}{\partial \theta_0} = \frac{1}{3}(-1.75 - 3.22 - 1.67) = -2.213$

Gradient for θ_1: $\frac{\partial J(\theta)}{\partial \theta_1} = \frac{1}{3}(-1.75 - 6.44 - 5.01) = -4.4$

Thus, $\theta_0 = 0 - \alpha(-2.213) = 2.213$ and $\theta_1 = 0 - \alpha(-4.4) = 4.4$

Rubric:

- 5 points for correctly finding the sub-terms.
- 5 points for correctly finding the solution.

2. If a loss function is non-convex and has multiple shallow local minimum values, should we use batch or stochastic gradient descent? 5 (CO1) (PO1)

Solution:

SGD's noisy gradient estimates introduce random perturbations or noise at each step. If a local minimum is shallow (its basin is narrow and the loss advantage over nearby regions is small), the noise may be large enough to kick the solution out of it, allowing exploration of potentially better regions.

Rubric:

- 5 points for correctly explanation.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Technology (CSE)

QUIZ #1 (SET-A) **SUMMER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. Suppose the newly discovered "Hantavirus" affects 1% of the population. A test for this disease is 95% accurate. If a person tests positive, what is the probability they actually have the disease? 5 (CO1) (PO1)

Solution:

Let D = has the disease and T = tests positive. $P(D) = 0.01$, $P(D') = 0.99$, $P(T|D) = 0.95$, $P(T|D') = 0.05$. Thus, $P(T) = P(T|D)P(D) + P(T|D')P(D') = 0.059$.

Using the Bayes' theorem, $P(D|T) = \frac{P(T|D)P(D)}{P(T)} = 0.161$

Rubric:

- 3 points for correctly finding the sub-terms.
- 2 points for correctly finding the solution.

2. If a loss function is non-convex and has multiple shallow local minimum values, should we use batch or stochastic gradient descent? 5 (CO1) (PO1)

Solution:

SGD's noisy gradient estimates introduce random perturbations or noise at each step. If a local minimum is shallow (its basin is narrow and the loss advantage over nearby regions is small), the noise may be large enough to kick the solution out of it, allowing exploration of potentially better regions.

Rubric:

- 5 points for correct explanation.

3. Why is feature normalization important before applying gradient descent with L2 regularization? 5 (CO1) (PO1)

Solution:

L2 regularization penalizes large weights equally. If features are on different scales, weights for small-scale features become artificially large just to compensate for the small feature values. The regularization term punishes these weights heavily, effectively penalizing that feature's importance unfairly.

Rubric:

- 5 points for correct explanation.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-SAMPLE) **SUMMER SEMESTER, 2024-2025**

DURATION: 25 MINUTES **FULL MARKS: 20**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. For a binary logistic regression model, 2 (CO1) (PO1)

$$h_\theta(x) = \sigma(\theta^T x),$$

which condition is equivalent to predicting class 1 using a threshold of 0.5?

A. $\theta^T x < 0$

B. $\theta^T x \geq 0$

C. $\theta^T x \geq 0.5$

D. $\theta^T x = 1$

> **Solution:**
> Since $\sigma(z) \geq 0.5$ if and only if $z \geq 0$, the model predicts class 1 exactly when $\theta^T x \geq 0$.

2. Why is binary cross-entropy preferred over mean squared error for standard logistic regression? 2 (CO1) (PO1)

A. It removes the need for a sigmoid function.

B. It guarantees zero training error.

C. It is the negative log-likelihood of the Bernoulli model and gives a convenient gradient.

D. It makes every decision boundary nonlinear.

> **Solution:**
> Binary cross-entropy follows from maximum likelihood under a Bernoulli output model. Together with the sigmoid, it simplifies the gradient to an error term of the form $(\hat{y} - y)x$.

3. Which expression is the vectorized gradient of the unregularized logistic regression loss for m examples, where $X \in \mathbb{R}^{m \times d}$? 2 (CO1) (PO1)

A. $\frac{1}{m} X(\hat{y} - y)$

B. $\frac{1}{m} X^T(\hat{y} - y)$

C. $\frac{1}{m}(\hat{y} - y)^T X^T$

D. $X^T X\theta$

Solution:

The prediction error vector has dimension m, so multiplying it by $X^T \in \mathbb{R}^{d\times m}$ produces a gradient vector in $\mathbb{R}^d$:

$$\nabla_\theta J(\theta) = \frac{1}{m} X^T(\hat{y} - y).$$

4. What happens if all hidden layers of a neural network use only affine transformations and no nonlinear activation functions? 2 (CO2) (PO1)

   A. The network can represent any continuous function.

   B. The entire network is equivalent to a single affine transformation.

   C. Backpropagation becomes mathematically impossible.

   D. The network automatically behaves like a nearest-neighbour classifier.

Solution:

A composition of affine transformations is still affine. Therefore, additional layers do not increase the expressive power unless nonlinear activation functions are inserted between them.

5. Which statement best explains why sigmoid activations can cause vanishing gradients in a deep network? 2 (CO2) (PO1)

   A. The sigmoid derivative is always greater than 1.

   B. The sigmoid output is unbounded.

   C. Its derivative is at most 0.25**, so repeated multiplication can rapidly shrink gradients.**

   D. It is not differentiable at the origin.

Solution:

Since

$$\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big) \le 0.25,$$

multiplying many such derivatives during backpropagation may make the gradient extremely small in earlier layers.

6. Which statement correctly describes the Universal Approximation Theorem? 2 (CO2) (PO1)

   A. A single neuron can exactly represent every continuous function.

   B. Gradient descent always finds the optimal approximating network.

   C. A one-hidden-layer network with enough neurons and a suitable nonlinear activation can approximate any continuous function on a compact domain arbitrarily well.

   D. Deep networks always require more parameters than shallow networks.

Solution:

The theorem is an existence result about representational capacity. It does not state how to find the parameters, how many neurons are needed, or whether the learned model will generalize.

7. A logistic regression model predicts $\hat{y} = 0.2$ for an example whose true label is $y = 1$. Compute the binary cross-entropy loss for this example. 2 (CO1) (PO1)

Solution:

For $y = 1$,

$$L = -\log(\hat{y}) = -\log(0.2) \approx 1.609.$$

Rubric:

- 1 mark for selecting the correct loss expression.
- 1 mark for the correct numerical value.

8. Briefly compare L1 and L2 regularization in terms of their effects on model weights. 2 (CO1) (PO1)

Solution:

L1 regularization adds a penalty proportional to $\sum_j |\theta_j|$ and can drive some weights exactly to zero, producing sparse models. L2 regularization adds a penalty proportional to $\sum_j \theta_j^2$ and generally shrinks all weights smoothly toward zero without making many of them exactly zero.

Rubric:

- 1 mark for correctly describing L1 regularization.
- 1 mark for correctly describing L2 regularization.

9. Explain why initializing all weights in a hidden layer to zero prevents the neurons in that layer from learning different features. 2 (CO2) (PO1)

Solution:

With identical zero initialization, all neurons in the layer produce identical outputs and receive identical gradients. Their weights therefore remain equal after every update. This symmetry prevents the neurons from learning distinct features.

Rubric:

- 1 mark for identifying identical outputs or gradients.
- 1 mark for explaining the resulting symmetry problem.

10. For a neural-network output layer using a sigmoid activation and binary cross-entropy loss, state the simplified expression for $dZ^{[L]}$ and explain why this simplification is useful. 2 (CO2) (PO1)

Solution:

The output-layer error is

$$dZ^{[L]} = A^{[L]} - Y.$$

The derivative terms from the sigmoid and the binary cross-entropy loss cancel, leaving a simple prediction-error expression. This makes backpropagation computationally efficient and easier to implement.

Rubric:

- 1 mark for the correct expression.
- 1 mark for a valid explanation of the simplification.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-A) **SUMMER SEMESTER, 2024-2025**

DURATION: 15 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. How can logistic regression produce a nonlinear decision boundary in the original input space? 3 (CO1) (PO1)

   A. By replacing gradient descent with Newton's method

   B. By adding polynomial features

   C. By adding the sigmoid function

   D. By setting all weights equal

> **Solution:**
>
> Logistic regression remains linear in its parameters, but nonlinear feature transformations such as: x_1^2, x_2^2, or $x_1 x_2$ can create nonlinear boundaries in the original feature space.

2. Why the does the universal approximation theorem not guarantee useful learning from finite data? 3 (CO1) (PO1)

   A. Approximation capacity does not mean successful optimization

   B. Approximation capacity is limited by the feature space

   C. The theorem prohibits linear functions with continuous output

   D. The theorem does not allow the model to rescale and shift normalized activations

> **Solution:**
>
> A network may have enough capacity to represent the target function, but training may not find the needed parameters, and fitting observed data does not ensure good performance on unseen examples.

3. How does batch normalization behave differently during training and inference? 3 (CO1) (PO1)

   A. The activation function changes at inference, making the gradients smoother

   B. Gradients are computed during inference and then saved in memory for further training

   C. Inference is not done in mini-batches, only sequentially

   D. Training uses mini-batch statistics, while inference uses stored estimates

Solution:

During training, batch mean and variance are used. During inference, running estimates provide stable deterministic normalization.

4. In gradient descent, a learning rate that is excessively large is most likely to cause: 3 (CO1) (PO1)

   A. Perfect generalization

   B. All parameters become zero

   C. Oscillation or divergence

   D. The loss to become convex

Solution:

A large learning rate can overshoot low-loss regions repeatedly and may cause the optimization process to diverge.

5. Logistic regression and conventional neural-network classifiers typically assume that each input instance belongs to exactly one class. How would you design a classification system for cases in which a single instance may belong to multiple classes simultaneously? 3 (CO2) (PO1)

Solution:

Use a multi-label classification model with one sigmoid output neuron per class, exactly like regular neural networks. Train the model using binary cross-entropy loss for each output, and assign every class whose predicted probability exceeds a chosen threshold.

Rubric:

- 1 mark for the correct formulation of ideas.
- 2 mark for a valid explanation.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-B) **SUMMER SEMESTER, 2024-2025**

DURATION: 15 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. A logistic-regression classifier uses a threshold τ instead of 0.5. Which decision boundary is correct? 3 (CO1) (PO1)

   A. $\theta^T x = \tau$

   B. $\theta^T x = log(\tau)$

   C. $\theta^T x = log(\tau/1 - \tau)$

   D. $\theta^T x = 1/(1 + e^{-\tau})$

   Solution:

   Setting $\sigma(\theta^T x) = \tau$ and solving for $\theta^T x$ gives us $\theta^T x = log(\tau)$

2. Why is the bias term often excluded from regularization? 3 (CO1) (PO1)

   A. The bias has no gradient

   B. Adding the bias makes the loss function non-convex

   C. The gradient of the bias is 1 as the features are normalized before training

   D. The bias only shifts the decision boundary

   Solution:

   Feature weights determine sensitivity to input variables, while the bias primarily shifts the boundary. It is therefore commonly left unpenalized.

3. Why can sigmoid activations contribute to vanishing gradients in deep networks? 3 (CO1) (PO1)

   A. The sigmoid derivative is always greater than one

   B. Its derivative is at most 0.5, so repeated multiplication can shrink gradients

   C. Sigmoid has no derivative at zero

   D. None of the above

   Solution:

   Its derivative is at most 0.25. Backpropagation multiplies many derivatives. Repeated factors no larger than 0.25 can make the gradient extremely small in early layers.

4. Why can momentum accelerate training in a long, narrow loss valley? 3 (CO1) (PO1)

   A. It removes all stochasticity from the gradient

   B. It accumulates velocity in consistent directions

   C. It guarantees the global minimum

   D. The loss to become convex if sigmoid is used alongside

Solution:

Momentum reinforces motion along directions where gradients agree over time and reduces zig-zagging across steep directions.

5. Logistic regression and conventional neural-network classifiers typically assume that each input instance belongs to exactly one class. How would you design a classification system for cases in which a single instance may belong to multiple classes simultaneously? 3 (CO2) (PO1)

Solution:

Use a multi-label classification model with one sigmoid output neuron per class, exactly like regular neural networks. Train the model using binary cross-entropy loss for each output, and assign every class whose predicted probability exceeds a chosen threshold.

Rubric:

- 1 mark for the correct formulation of ideas.
- 2 mark for a valid explanation.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Technology (CSE)

QUIZ #3 (SAMPLE) **SUMMER SEMESTER, 2024-2025**

DURATION: 15 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. Which statement correctly distinguishes the Skip-gram and CBOW objectives used in Word2Vec? 3 (CO1) (PO1)

   A. Skip-gram predicts a center word from its surrounding words, whereas CBOW predicts surrounding words from the center word.

   B. Skip-gram predicts surrounding words from a center word, whereas CBOW predicts the center word from its surrounding words.

   C. Skip-gram uses one-hot vectors, whereas CBOW does not use vector representations.

   D. Skip-gram is used only for classification, whereas CBOW is used only for regression.

Solution:

Skip-gram receives a center word and attempts to predict words occurring in its context window. CBOW performs the reverse operation: it combines the surrounding context and predicts the center word.

2. Why can a standard recurrent neural network struggle to learn dependencies between words that are far apart in a sequence? 3 (CO1) (PO1)

   A. The hidden state cannot contain any numerical values.

   B. Recurrent networks do not share parameters across time steps.

   C. During backpropagation through many time steps, gradients may repeatedly shrink or grow, causing vanishing or exploding gradients.

   D. A recurrent network processes every token independently and has no hidden state.

Solution:

Training an RNN requires gradients to pass through the recurrent computation repeatedly. Multiplication by many Jacobian terms may make the gradient approach zero or become excessively large. This makes long-range dependencies difficult to learn.

3. What is the primary purpose of positional encoding in a Transformer? 3 (CO1) (PO1)

   A. To reduce the vocabulary size before applying softmax

   B. To ensure that every token has the same embedding

   C. To provide information about token order because self-attention alone does not inherently encode sequence position

D. To prevent the decoder from accessing the encoder output

Solution:

Self-attention compares tokens without inherently knowing their order. Positional encodings are added to token embeddings so that the model can distinguish, for example, the first token from the fifth token and learn order-dependent relationships.

4. A classification-tree node contains 8 examples: 4 from class A and 4 from class B. A candidate split creates: 3 (CO2) (PO1)

   - left child: $4A$, $0B$;
   - right child: $0A$, $4B$.

   Using entropy with base-2 logarithms, what is the information gain of this split?

   A. 0

   B. 0.5

   C. 1

   D. 2

Solution:

The parent entropy is

$$H_{\text{parent}} = -\tfrac{1}{2}\log_2 \tfrac{1}{2} - \tfrac{1}{2}\log_2 \tfrac{1}{2} = 1.$$

Both children are pure, so each has entropy 0. Therefore,

$$IG = 1 - \left(\tfrac{4}{8}\cdot 0 + \tfrac{4}{8}\cdot 0\right) = 1.$$

5. A language model uses the following one-dimensional vectors for a single attention head: 3 (CO2) (PO1)

$$q = [1], \qquad k_1 = [0], \qquad k_2 = [\ln 3], \qquad v_1 = [2], \qquad v_2 = [8].$$

Assume $d_k = 1$. Compute the attention output for q, and briefly explain what the result means.

Solution:

The attention logits are

$$qk_1^T = 0, \qquad qk_2^T = \ln 3.$$

Thus,

$$\text{softmax}([0, \ln 3]) = \left[\frac{e^0}{e^0 + e^{\ln 3}}, \frac{e^{\ln 3}}{e^0 + e^{\ln 3}}\right] = \left[\frac{1}{4}, \frac{3}{4}\right].$$

The output is the weighted sum

$$\frac{1}{4}(2) + \frac{3}{4}(8) = \frac{1}{2} + 6 = 6.5.$$

The second token receives greater attention because its key is more compatible with the query; consequently, its value contributes more strongly to the output.

Rubric:

- 1 mark for obtaining the logits and attention weights.
- 1 mark for computing the weighted value.
- 1 mark for correctly interpreting the larger contribution of the second value.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

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Department of Computer Science and Technology (CSE)

QUIZ #3 (SET-B) **SUMMER SEMESTER, 2024-2025**

DURATION: 15 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 11 (eleven)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. Decision trees are described as a *non-parametric* method. What does this mean? 1 (CO1) (PO1)

   A. The number of parameters is not fixed

   B. The model has zero learnable parameters of any kind

   C. The tree can only be used for classification

   D. The test function at every node must be linear in all input features simultaneously

Solution:

Non-parametric methods do not assume a fixed number of parameters set before training. A decision tree's size (number of splits, nodes, and leaves) is determined by the data itself and can keep growing as needed, unlike a parametric model such as logistic regression, which has a fixed number of parameters regardless of dataset size.

2. In a *univariate* decision tree, each internal node's test function thresholds only a single input feature (e.g., $x_1 > w_{10}$). What shape do the resulting decision boundaries take in the input space? 1 (CO1) (PO1)

   A. Smooth, arbitrarily curved boundaries that can take any shape

   B. A single hyperplane at an arbitrary angle to the axes

   C. Circular boundaries centered on the training data's mean

   D. Axis-aligned boundaries, partitioning the space into rectangular regions

Solution:

Since each test compares a single feature to a threshold, every individual decision boundary is a line (or hyperplane) perpendicular to that feature's axis. Composing many such splits partitions the input space into axis-aligned rectangular regions, as shown in the two-feature decision-region example from lecture.

3. Hierarchical decisions can locate a region efficiently, needing on the order of $\log(b)$ decisions in the best (balanced) case. What does b represent here? 1 (CO1) (PO1)

   A. The total number of training examples in the dataset

   B. The depth of the shallowest possible tree, regardless of the number of regions

   C. The number of leaves the input space is partitioned into

D. The number of features in the input vector

Solution:

b is the number of regions (leaves) the space is divided into. A balanced hierarchical structure needs only about $\log(b)$ sequential test decisions to route an input to its region, since each decision roughly halves the number of remaining candidate regions.

4. Both entropy and Gini impurity are examples of an impurity measure $\phi(p, 1-p)$, where p is the proportion of one class in a node. Which property do BOTH measures share by definition? 1 (CO1) (PO1)

   A. They equal 0 for a pure node and reach their maximum at a 50/50 class mixture

   B. They equal 1 for a pure node and 0 at a 50/50 mixture

   C. They only apply when there are exactly two classes and cannot be extended further

   D. They increase monotonically as p goes from 0 to 1

Solution:

By definition, an impurity measure is 0 when a split is pure (every example in one class) and rises to its maximum at the most balanced, 50/50, mixture, then falls again as one class begins to dominate again. This "rises then falls" shape holds for both entropy and Gini impurity.

5. Unlike an image, which always has a fixed $H \times W \times C$ shape, a sentence has variable length. What specific difficulty does this create when trying to represent an entire sentence? 1 (CO1) (PO1)

   A. A sentence cannot be split into individual vectors while preserving the semantics

   B. A sentence always requires more storage than an image of any resolution due to its variable size

   C. A sentence's meaning becomes completely independent of word order

   D. A sentence cannot simply be flattened into a single vector of fixed size

Solution:

Because sentences vary in length, they cannot be flattened into a vector of fixed, predetermined size the way a fixed-shape image can. This is one reason architectures built for sequences must be able to handle variable-length input, unlike architectures designed for fixed-size images.

6. Convolutional sliding windows use a fixed receptive field. Why does this make convolution a poor fit for capturing dependencies in language? 1 (CO1) (PO1)

   A. A fixed receptive field can only look at a local span of words

   B. Convolution can only be applied to inputs of a single fixed length, so it cannot process sentences at all

   C. A fixed receptive field forces every word in the vocabulary to share an identical embedding

D. Convolutional filters can only be applied to images, not sentences

Solution:

A convolutional filter with a fixed receptive field only ever looks at a small, local window of neighboring words at a time. Since relevant dependencies in language can span arbitrarily long distances (well beyond the filter's fixed window), a fixed receptive field cannot, by itself, capture such long-range relationships without stacking many layers or otherwise extending its effective reach.

7. Training methods such as Skip-gram and CBOW are described as *self-supervised learning*. What makes them self-supervised rather than standard supervised learning? 1 (CO1) (PO1)

A. A human annotator labels every word with its correct embedding vector before training begins

B. The model receives no training signal of any kind and updates its weights randomly

C. Training pairs are automatically generated from unlabeled text itself

D. The vocabulary is manually clustered into semantic categories before training

Solution:

The "labels" used for training (e.g., which words appear in the context window around a given center word) come directly from the structure of ordinary, unannotated text – no human needs to manually tag anything. This automatic creation of supervision signal from raw data is exactly what defines self-supervised learning.

8. A node contains 15 examples: 9 of class A and 6 of class B. A candidate split creates: left child: $(5A, 1B)$, right child: $(4A, 5B)$. Using Gini impurity, what is the gain from this split? 1 (CO1) (PO1)

A. 0.02

B. 0.07

C. 0.12

D. 0.17

Solution:

Parent Gini: $1 - \left(\frac{9}{15}\right)^2 - \left(\frac{6}{15}\right)^2 = 1 - (0.36 + 0.16) = 0.48.$

Left child Gini: $1 - \left(\frac{5}{6}\right)^2 - \left(\frac{1}{6}\right)^2 = 1 - (0.694 + 0.028) = 0.28.$

Right child Gini: $1 - \left(\frac{4}{9}\right)^2 - \left(\frac{5}{9}\right)^2 = 1 - (0.198 + 0.309) = 0.49.$

Weighted post-split Gini: $\frac{6}{15}(0.28) + \frac{9}{15}(0.49) = 0.4(0.28) + 0.6(0.49) = 0.112 + 0.294 = 0.41.$

Gain $= 0.48 - 0.41 = 0.07.$

9. Which RNN input-output topology best matches the task of generating a full image caption (a sequence of words) from a single input image? 1 (CO1) (PO1)

A. Many-to-one
B. One-to-one
C. Many-to-many
D. One-to-many

Solution:

A single, fixed input (the image) is used to generate a variable-length sequence of outputs (the caption's words), which matches the one-to-many topology: one input followed by a sequence of dependent outputs.

10. Which of the following is a genuine limitation of the greedy, information-gain-based split strategy used in decision tree induction? 1 (CO1) (PO1)

A. It cannot be applied to attributes with numeric values due to infinite possibilities
B. A locally optimal split can lead to a globally worse tree due to no backtracking
C. It always produces the tree with the fewest possible leaf nodes for minimum spanning examples
D. It produces the minimum-depth tree for any dataset

Solution:

Greedy search commits to the best split *at the current node* and cannot revisit that decision later, even if a different split would have led to a better overall tree.

11. The classic sequence-to-sequence (encoder-decoder) architecture suffers from a "bottleneck problem," evidenced empirically by BLEU scores dropping sharply as sentence length increases. Explain: 5 (CO2) (PO1)

- Why does this bottleneck occur?
- How does the attention mechanism removes the need for a single fixed-size context vector?

Solution:

(a) The encoder compresses the *entire* input sequence into one fixed-size context vector regardless of sentence length. As the sentence grows, this fixed-capacity representation must encode more information, so earlier or fine-grained details get overwritten or lost, degrading translation quality.

(b) Attention lets the decoder, at each output step, directly access *all* encoder hidden states rather than a single summary vector. It computes a relevance-weighted combination of every encoder state (via query-key comparisons) specific to the current decoding step, so no single vector has to carry the whole sentence's meaning.

Rubric:

- 2.5 marks for a correct, precise explanation of why the fixed-size vector causes degradation with length.
- 2.5 marks for correctly explaining that attention replaces the single vector with a per-step, weighted combination over all encoder states.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Technology (CSE)

QUIZ #3 (SET-A) **SUMMER SEMESTER, 2024-2025**

DURATION: 15 MINUTES **FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 11 (eleven)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

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1. A node (N=12) is split into three branches of sizes 5, 4, 3 with entropies 0.72, 0.99, and 0 respectively. If the parent entropy is 0.98, what is the information gain? 1 (CO1) (PO1)

   A. 0.30

   B. 0.35

   C. 0.40

   D. 0.45

   Solution:

   Weighted entropy $= \frac{5(0.72)+4(0.99)+3(0)}{12} = \frac{7.56}{12} = 0.63$. Gain $= 0.98 - 0.63 = 0.35$.

2. A root node has entropy 0.94. Attribute P, Q, R give weighted post-split entropies 0.693, 0.788, 0.911. Which is chosen, and why? 1 (CO1) (PO1)

   A. P, since it yields the highest information gain

   B. R, since higher post-split entropy signals more informative branches

   C. Q, since its gain is closest to the mean gain of all attributes

   D. Any of them, since gain does not depend on post-split entropy

   Solution:

   Gain $=$ 0.94 $-$ (post-split entropy); the lowest post-split entropy (P) gives the highest gain (0.247).

3. Which of the following is a genuine limitation of the greedy, information-gain-based split strategy used in decision tree induction? 1 (CO1) (PO1)

   A. It cannot be applied to attributes with numeric values due to infinite possibilities

   B. A locally optimal split can lead to a globally worse tree due to no backtracking

   C. It always produces the tree with the fewest possible leaf nodes for minimum spanning examples

   D. It produces the minimum-depth tree for any dataset

Solution:

Greedy search commits to the best split *at the current node* and cannot revisit that decision later, even if a different split would have led to a better overall tree.

4. Which of the following statements about one-hot word encoding is **FALSE**? 1 (CO1) (PO1)

   A. The cosine similarity between any two distinct one-hot vectors is zero

   B. The dimensionality of the encoding grows linearly with vocabulary size

   C. One-hot vectors are extremely sparse, with only one non-zero entry

   D. One-hot vectors encode semantic similarity between related words

Solution:

One-hot vectors are mutually orthogonal by construction and carry no notion of semantic similarity; this motivates learned dense embeddings.

5. The relation king − man + woman ≈ queen is often used to illustrate learned word embeddings. What does this analogy actually demonstrate? 1 (CO1) (PO1)

   A. That embeddings encode exact, rule-based grammatical gender agreement

   B. That relationships can emerge as roughly consistent vector offsets in the embedding space

   C. That each learned embedding dimension corresponds to a single attribute about a word

   D. That Word2Vec requires labeled analogy pairs during training to learn this behavior

Solution:

Training on co-occurrence statistics alone (self-supervised, no analogy labels) causes semantically related directions to emerge approximately as consistent vector offsets, not exact symbolic rules.

6. What is the primary motivation for sharing the same weight matrices (W_{hh}, W_{xh}, W_{hy}) across every time step of an RNN? 1 (CO1) (PO1)

   A. To keep the parameter count independent of sequence length

   B. To allow every time step to be computed in parallel, similar to convolution

   C. To act as an implicit regularizer that directly prevents overfitting

   D. To guarantee gradient stabilization (neither vanish nor explode) during training

Solution:

Reusing one transformation at every step avoids needing a distinct parameter set per sequence length; RNNs remain inherently sequential and weight sharing gives no gradient-stability guarantee.

7. Why does the *additive* cell-state update path in an LSTM help mitigate the vanishing gradient problem, compared to a vanilla RNN's fully multiplicative hidden-state recurrence? 1 (CO1) (PO1)

   A. Because addition operations block gradient flow entirely, so no vanishing can occur
   B. Because the cell state uses a ReLU activation, which never saturates
   C. Because the forget gate is mathematically constrained to always output 1
   D. None of the above

Solution:

Gradients through an addition are passed through largely unattenuated, unlike the repeated multiplicative recurrence in a vanilla RNN, which is why the cell state acts as a more stable pathway for gradient flow.

8. At a given time step, an LSTM's forget gate outputs values near 0 while its input gate outputs values near 1. What does this imply about the cell's behavior at that step? 1 (CO1) (PO1)

   A. It retains all prior memory and ignores the new candidate information
   B. It retains prior memory and writes in the new candidate information
   C. It discards prior memory and writes in the new candidate information
   D. The hidden state becomes numerically identical to the cell state

Solution:

A forget gate near 0 erases most of c_{t-1}; an input gate near 1 writes in nearly all of the new candidate values – together these replace old memory with new information.

9. Why can a standard recurrent neural network struggle to learn dependencies between words that are far apart in a sequence? 1 (CO1) (PO1)

   A. The hidden state cannot retain any numerical values
   B. Recurrent networks do not share parameters across time steps
   C. Gradients may repeatedly shrink or grow, causing vanishing or exploding gradients
   D. A recurrent network processes every token independently regardless of sentence length

Solution:

Training an RNN requires gradients to pass through the recurrent computation repeatedly. Multiplication by many Jacobian terms may make the gradient approach zero or become excessively large. This makes long-range dependencies difficult to learn.

10. What is the core motivation for multi-head attention? 1 (CO1) (PO1)

    A. A single learned subspace may capture some token relationships well and others poorly

B. The model must identify and keep only the single best-scoring head at inference time and it may change from sample to sample

C. Only the "Good" subspace receives gradient updates during backpropagation while not updating the "Bad" heads

D. Poorly performing subspaces are removed by the multi-headed self-attention mechanism

Solution:

Different linear projections emphasize different relationships between tokens; some are useful ("Good"), some are not ("Bad"). Multiple heads let the model learn several such subspaces simultaneously instead of relying on one.

11. The classic sequence-to-sequence (encoder-decoder) architecture suffers from a "bottleneck problem," evidenced empirically by BLEU scores dropping sharply as sentence length increases. Explain: 5 (CO2) (PO1)

- Why does this bottleneck occur?
- How does the attention mechanism removes the need for a single fixed-size context vector?

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